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Avadi, Chennai

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 28-04-2026

SMART CAMPUS EVENT MANAGEMENT SYSTEM

SDG: SDG 4 – Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- Managing campus events manually often leads to issues such as missed registrations, poor communication, and difficulty in tracking participant details.
- Colleges regularly conduct workshops, seminars, and technical events that require an organized digital platform.
- This project is important because it simplifies event creation, registration, and management for both students and administrators.
- Students can easily browse upcoming events, register online, and provide feedback through a user-friendly interface.
- Administrators can efficiently create, update, delete, and monitor event records in real time.
- The system also provides registration statistics and filtered event search for better decision-making.
- Its real-world relevance lies in automating university event workflows and reducing manual administrative effort.
- Overall, it is a full-stack web application built using Spring Boot, JPA, and Thymeleaf for smart campus management.

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PROBLEM STATEMENT



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-Many institutions still manage campus events using manual methods such as spreadsheets, emails, or notice boards.

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-This leads to inefficiencies in handling event information, registrations, and participant tracking.

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-Students often face difficulty in accessing updated event details and registering on time.

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-Administrators struggle with maintaining accurate records and managing multiple events simultaneously.

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-Lack of a centralized system results in poor communication and data inconsistency.

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-There is no efficient way to analyze participation or generate event statistics.

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-Manual processes increase the chances of errors and data redundancy.

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-Hence, there is a need for a reliable, automated system to streamline campus event management.

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OBJECTIVE(S)



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-To design and develop a centralized Smart Campus Event Management System for efficient event creation, registration, and tracking.

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-To implement automated event management with secure admin control and real-time access for students to improve accuracy and performance.

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PROPOSED SYSTEM



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- Develop a web-based Smart Campus Event Management System using Spring Boot and Thymeleaf for centralized access.

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- Provide separate modules for students and admins to ensure structured and role-based functionality.

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- Enable students to browse events, register online, and track their participation easily.

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- Allow admins to create, update, delete, and manage events efficiently through a secure dashboard.

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- Integrate REST APIs and JPA for seamless data handling and database operations.

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- Implement validation and authentication to ensure data accuracy and system security.

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- Include search filters and analytics to quickly find events and monitor participation statistics.

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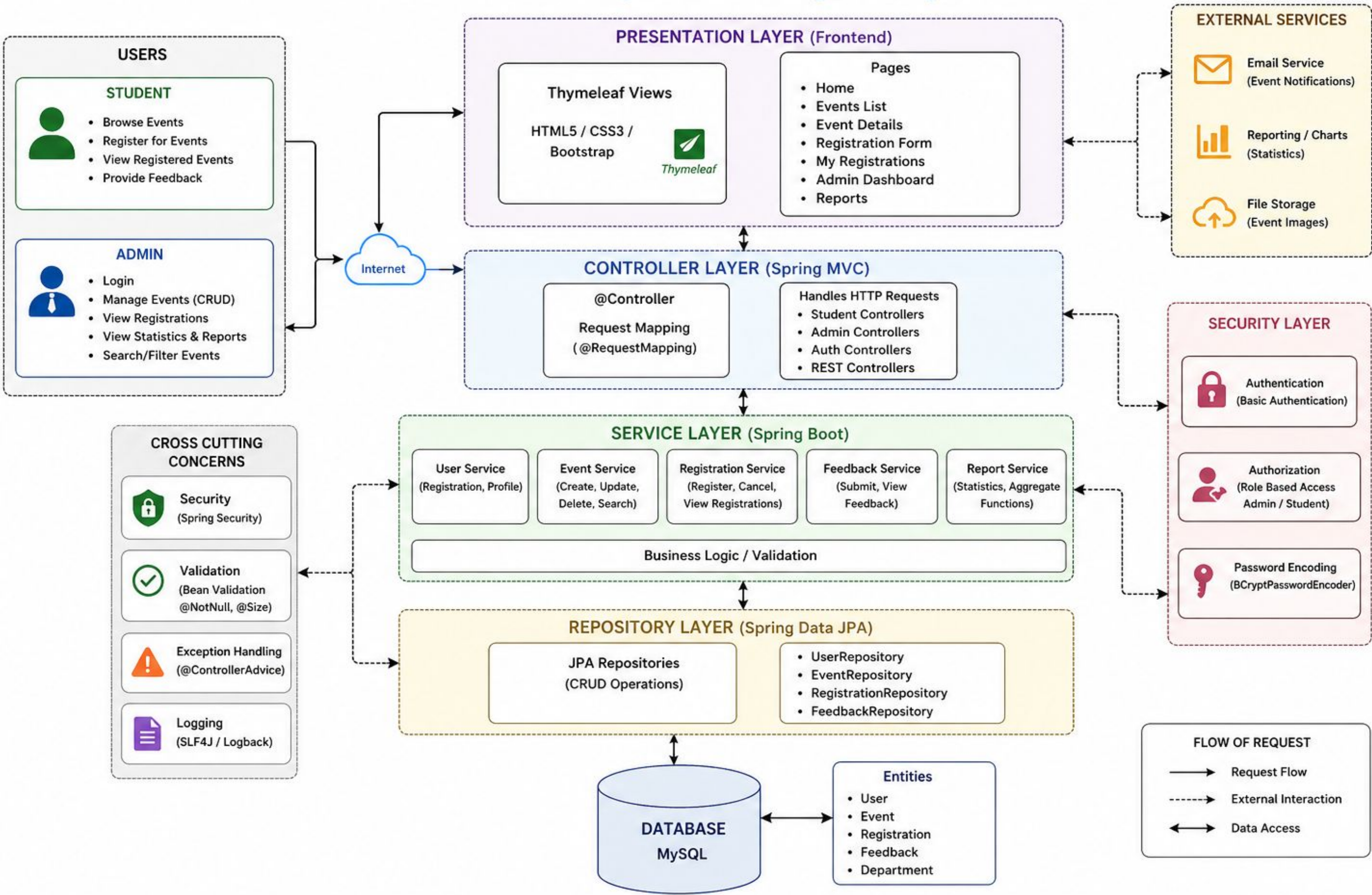
- Compared to manual systems, it reduces errors, saves time, improves accessibility, and ensures better data consistency.

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PROPOSED SYSTEM ARCHITECTURE

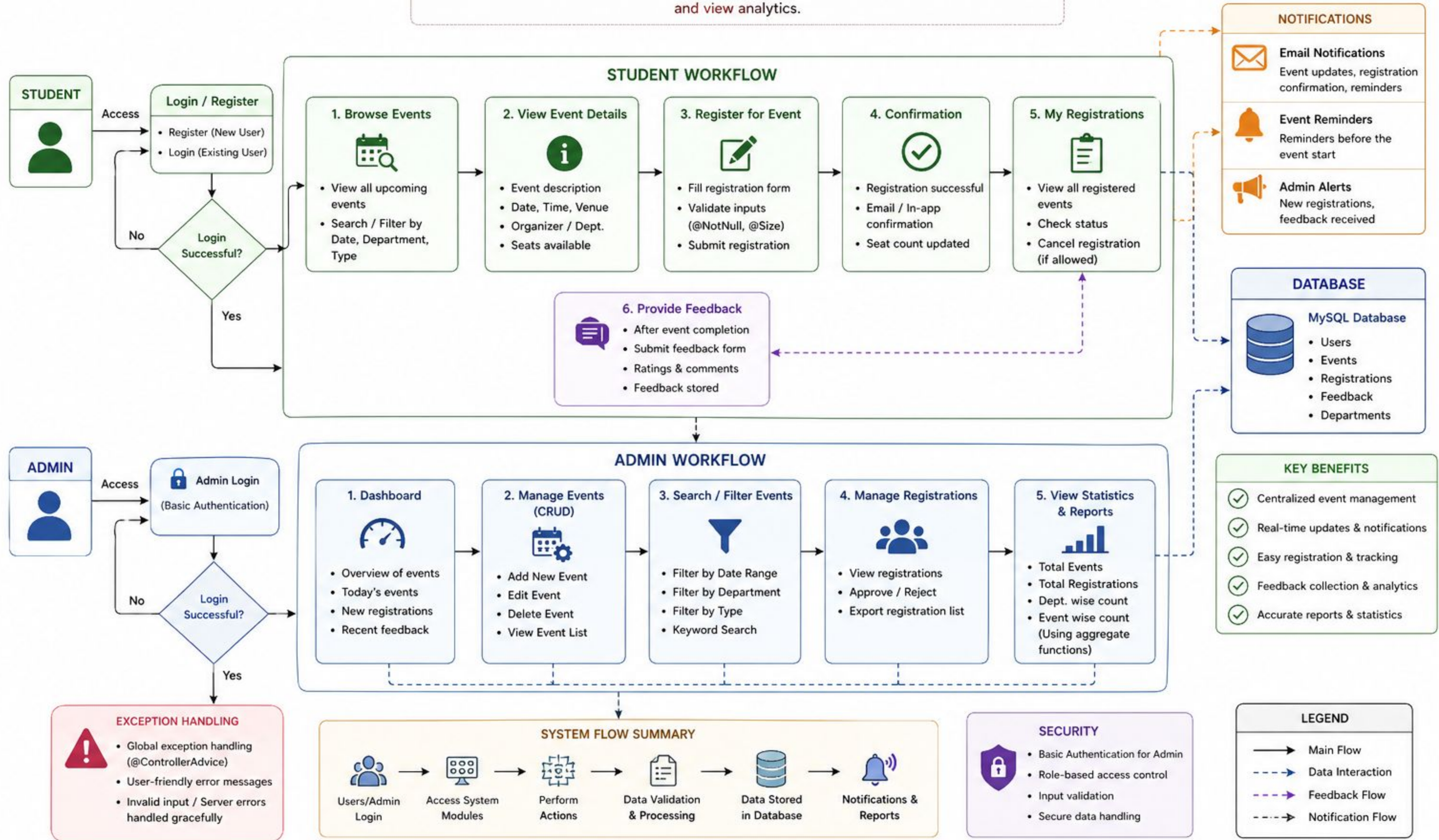
Smart Campus Event Management System



SMART CAMPUS EVENT MANAGEMENT SYSTEM

WORKFLOW DIAGRAM

The system allows students to view and register for events, and admins to manage events and view analytics.



TECHNOLOGIES USED



- **Frontend:**

HTML5, CSS3, Thymeleaf (for dynamic server-side rendering)

- **Backend:**

Java, Spring Boot (Spring MVC, Spring Data JPA, REST APIs)

- **Database:**

MySQL (Relational Database for storing events, users, registrations)

- **Tools:**

Spring Tool Suite (STS) / IntelliJ IDEA, Postman (API testing), Git & GitHub (version control), Maven (build tool)

- **Hardware/Software Requirements:**

Minimum 8 GB RAM, Intel i5 or above processor, Java JDK 8+, MySQL Server, Web Browser (Chrome/Edge), Operating System (Windows/Linux/Mac)

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MODULE DESCRIPTION



Module 1: Student Management

Handles student interactions such as browsing events, registering for events, and viewing registered event details.

Module 2: Admin Management

Manages admin functionalities including adding, updating, deleting events, and controlling event-related operations securely.

Module 3: Event Management

Implements core logic for event creation, scheduling, categorization, and search using filters like date, department, and type.

Module 4: Database Management

Handles data storage and retrieval using JPA, ensuring data consistency for events, users, and registrations.

Module 5: Analytics & Reporting

Generates event participation statistics and insights to help admins analyze trends and improve future event planning.

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IMPLEMENTATION SCREENSHOTS



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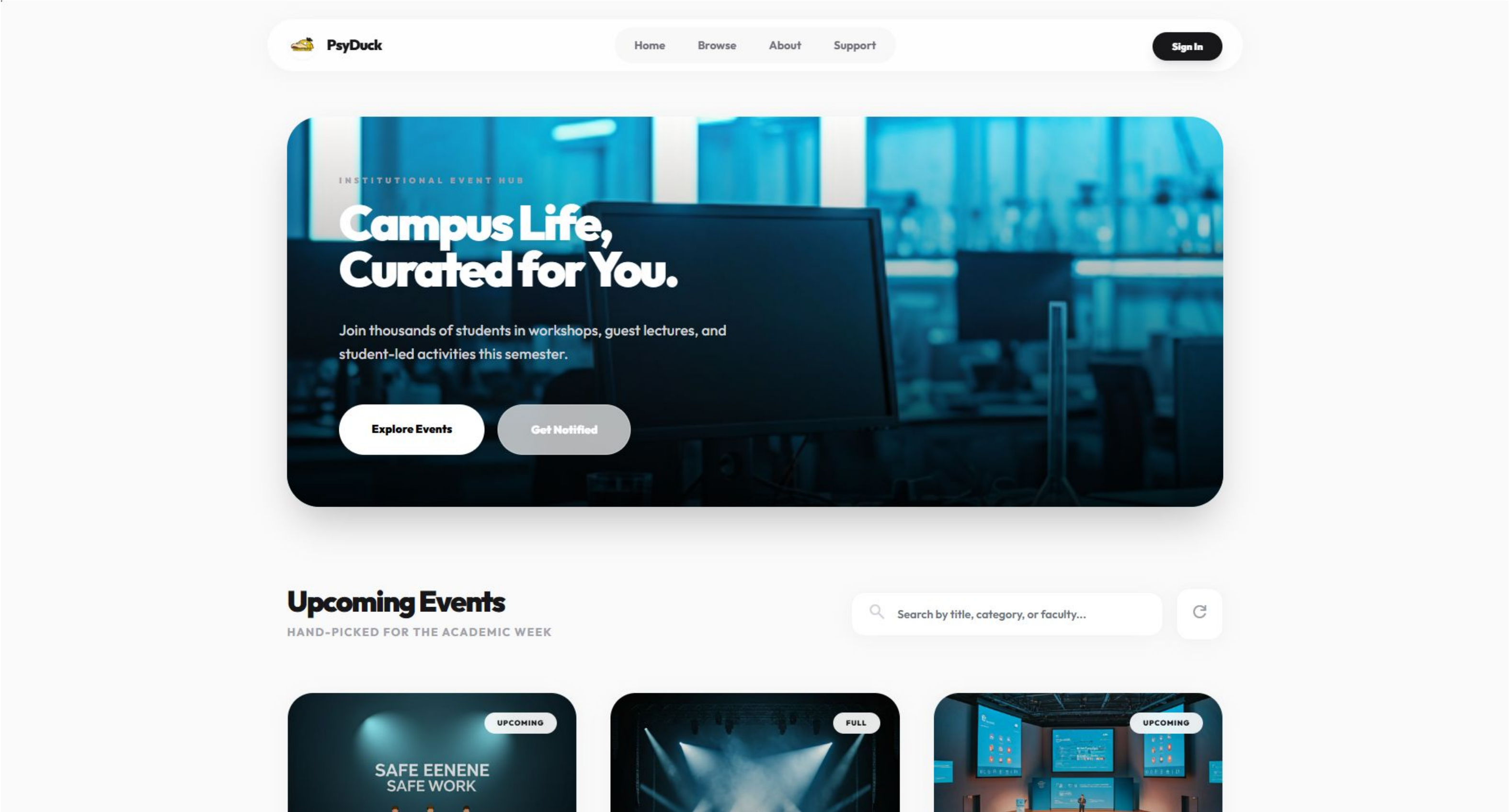
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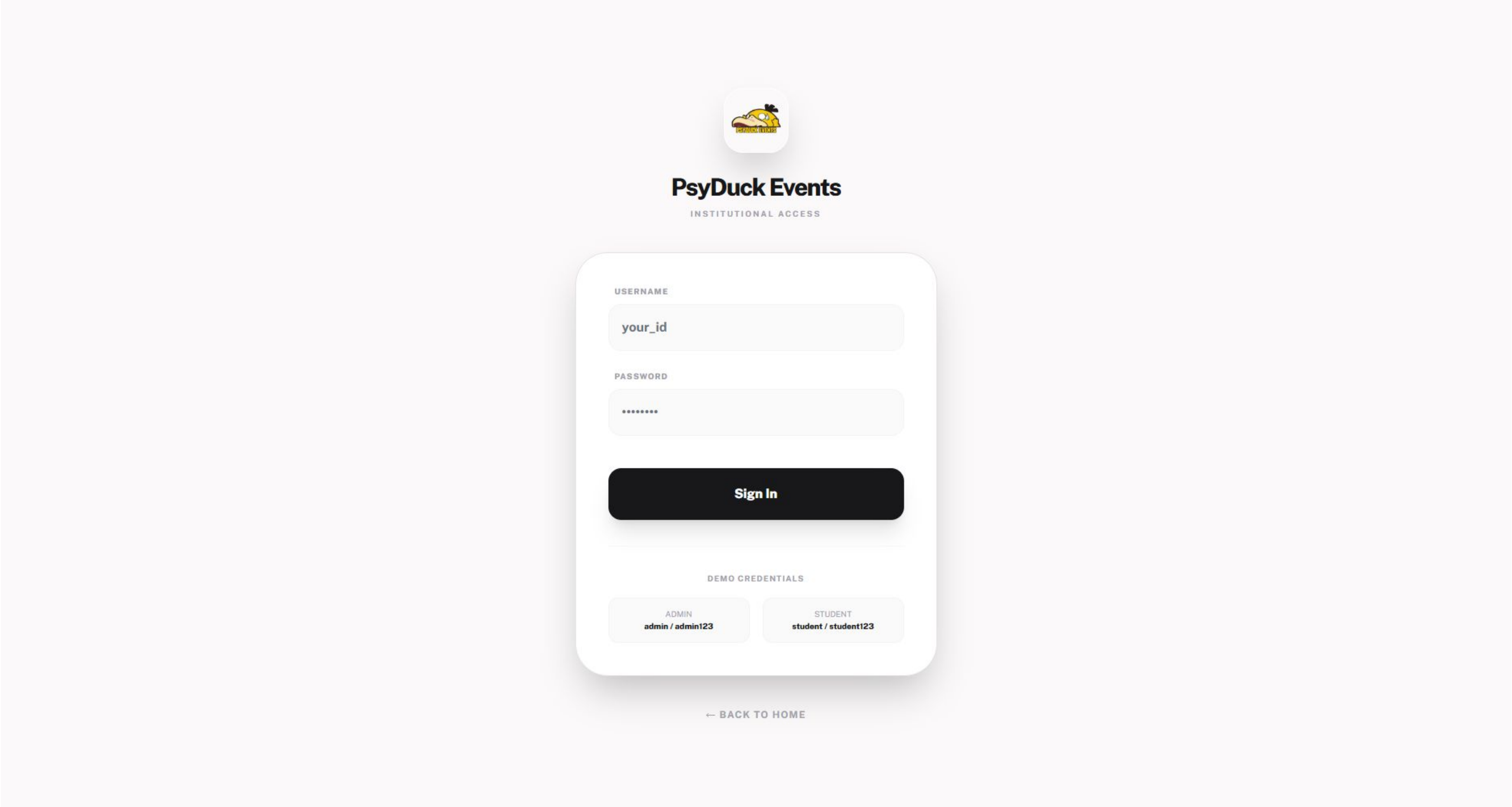
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IMPLEMENTATION SCREENSHOTS



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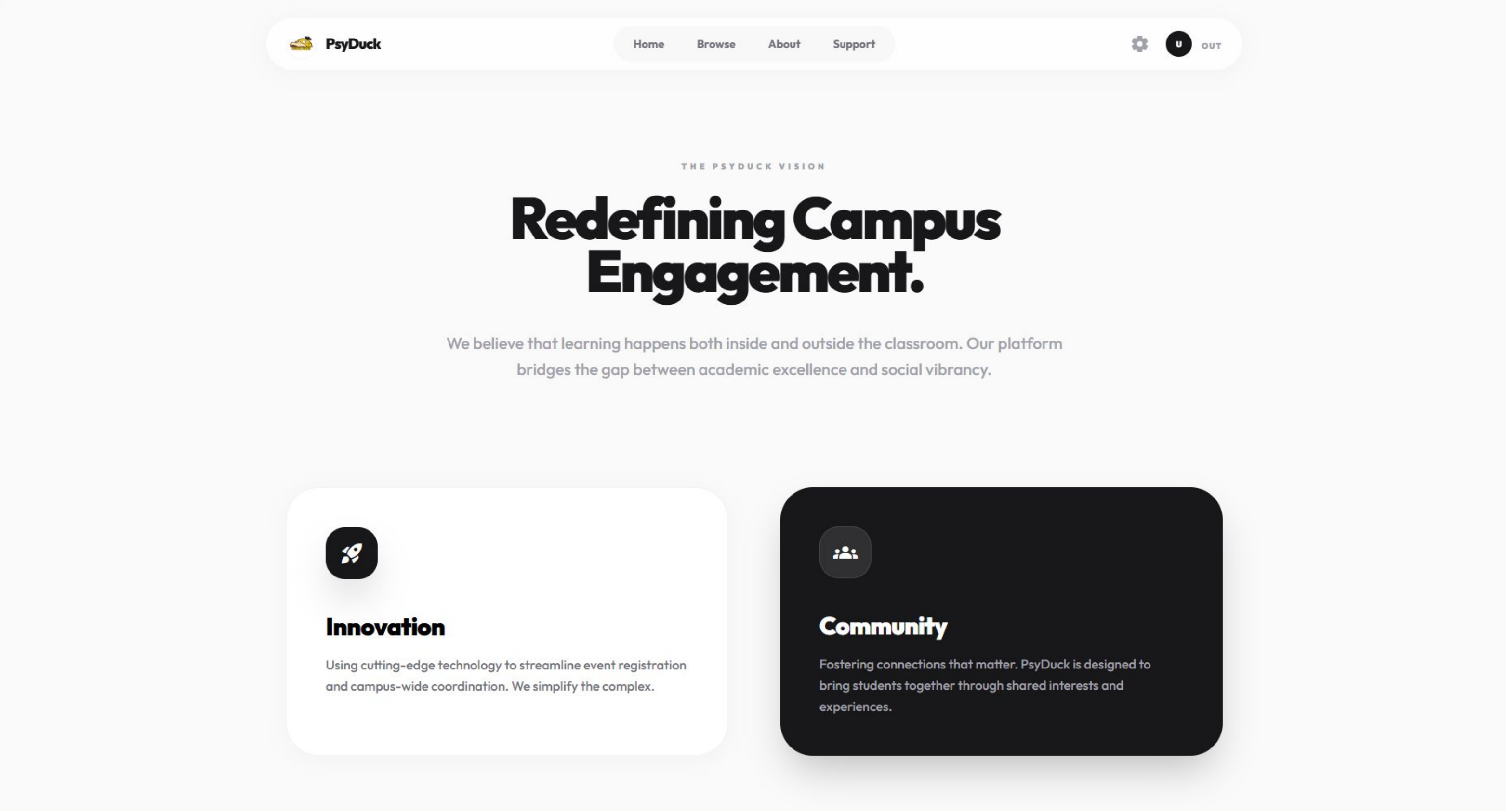
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IMPLEMENTATION SCREENSHOTS



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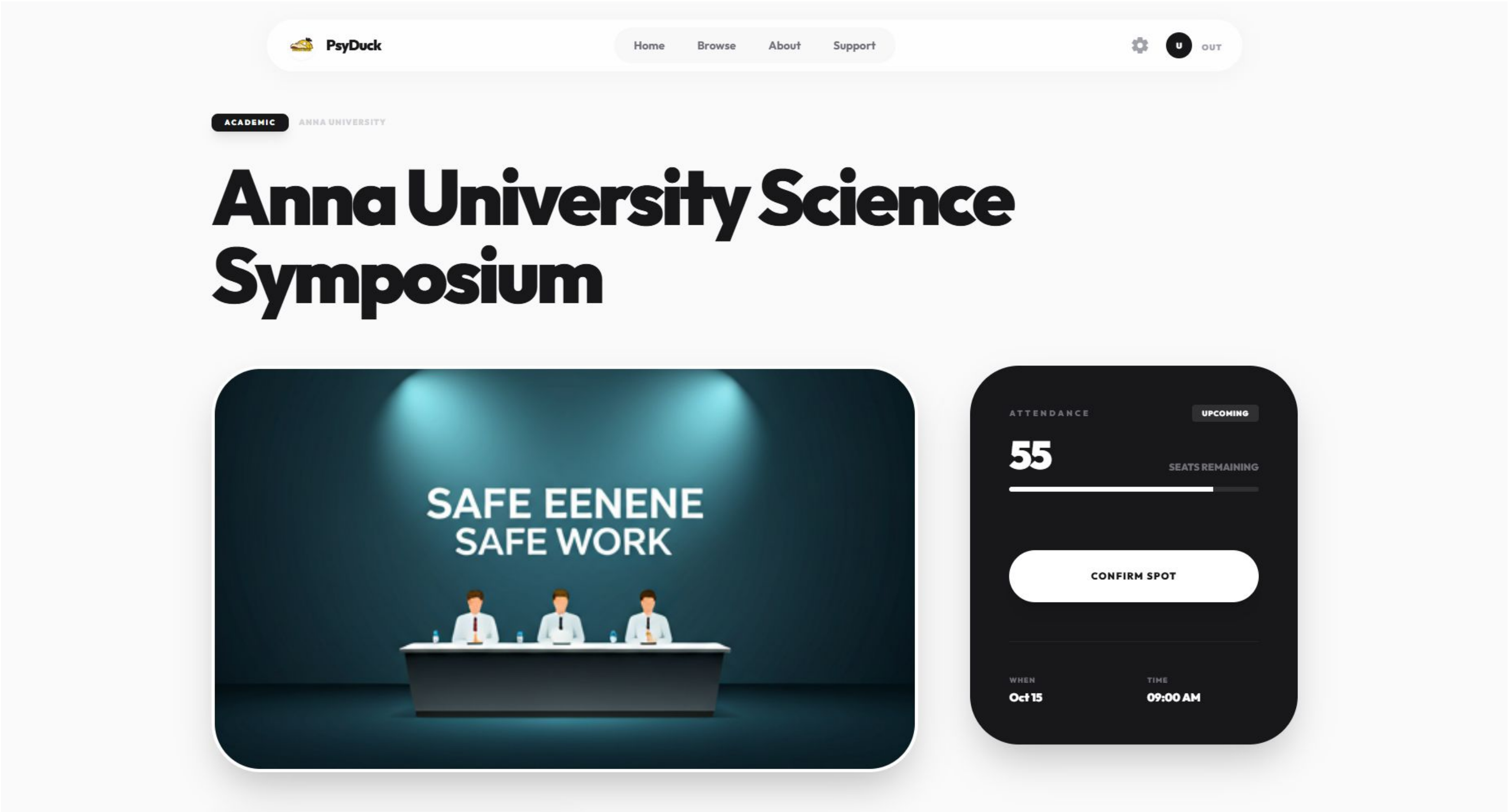
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ADVANTAGES & LIMITATIONS



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Advantages

- Centralized system for managing all campus events efficiently
- Reduces manual work and minimizes human errors
- Easy access for students to view and register for events anytime
- Secure admin control with authentication and role-based access
- Real-time updates and better communication of event information
- Supports search, filtering, and event analytics for better decision-making

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Limitations

- Requires internet access for full functionality
- Basic security (authentication only) may not handle advanced threats
- Scalability may be limited without cloud deployment
- Depends on proper data entry by admins for accuracy
- Initial setup and maintenance effort required

Thank you

